

TE Curve Verification Pipeline Best Model Collage Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report compares representative `TE Curve Verification Pipeline` TE-curve predictions for the current best reference, RCIM Model-Bank Reproduction, Wave 1 directional, and Wave 1 global models. Each model is shown as one four-image collage so local oscillation tracking can be inspected directly.

Scope

- each collage contains four deterministic held-out test curves;
- forward models are shown on forward curves only;
- backward models are shown on backward curves only;
- global Wave 1 models are shown on two forward and two backward curves;
- `Measured TE` uses the same line width as predictions and a dark-gray color for balanced visual comparison.

Metrics Summary

Backward Accepted Non PINN Incumbent Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
<code>accepted_periodic_gru_sequence_bw</code>	<code>accepted_non_pinn_incumbent</code>	Bw	0.001837	0.002196	3.494
<code>accepted_periodic_mlp_harmonic_bw</code>	<code>accepted_non_pinn_incumbent</code>	Bw	0.001912	0.002263	3.581

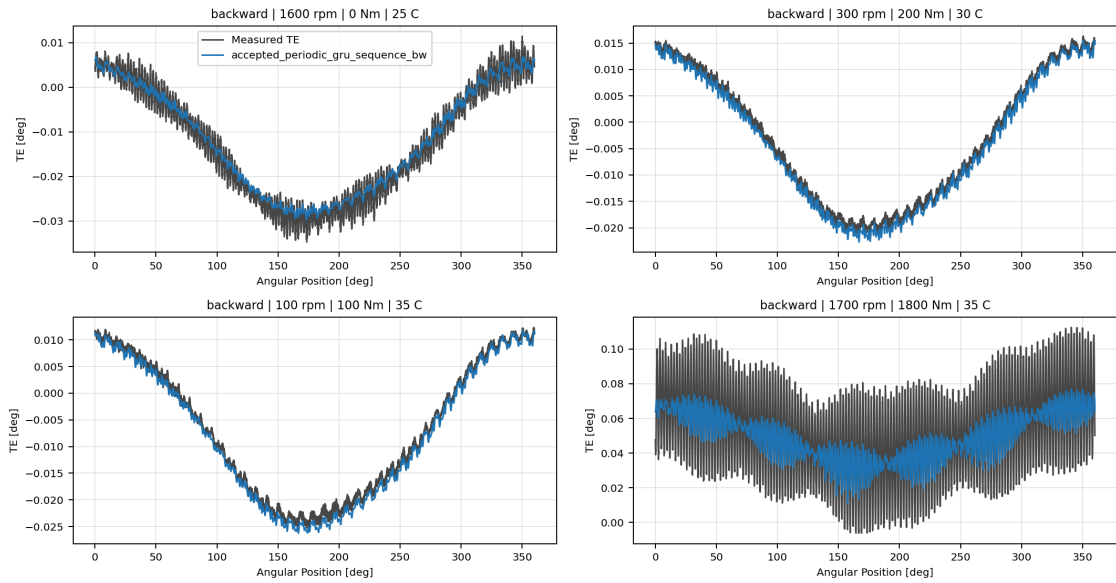
Backward Wave52r Offline Leader Cross Surface Promotion Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
<code>wave52r_promotion_k01_bw_seed_271828</code>	<code>wave52r_offline_leader_cross_surface_promotion</code>	Bw	0.001587	0.001873	3.018
<code>wave52r_promotion_h08_bw_seed_161803</code>	<code>wave52r_offline_leader_cross_surface_promotion</code>	Bw	0.001933	0.002287	3.710

Collage Gallery - Backward Accepted Non PINN Incumbent Models

accepted_periodic_gru_sequence_bw

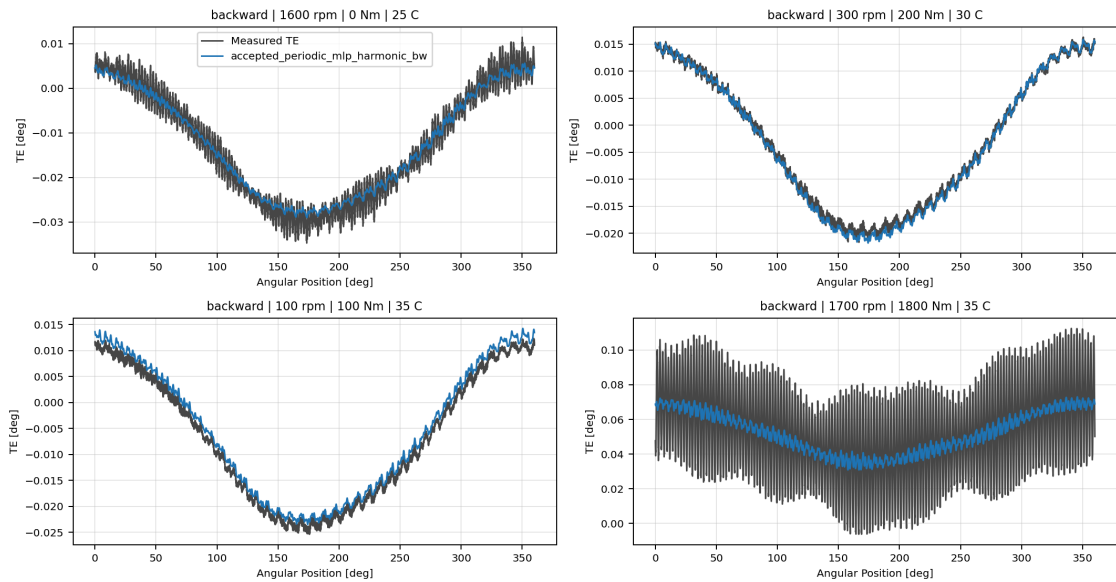
accepted_periodic_gru_sequence_bw



accepted_periodic_gru_sequence_bw TE Curve Verification Pipeline collage

accepted_periodic_mlp_harmonic_bw

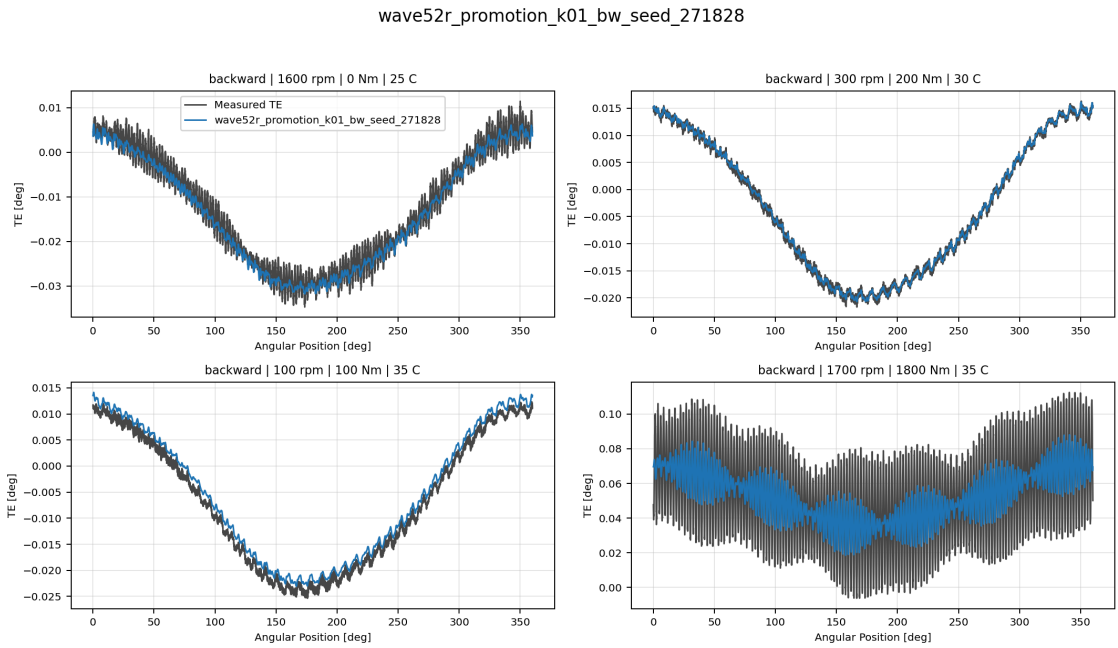
accepted_periodic_mlp_harmonic_bw



accepted_periodic_mlp_harmonic_bw TE Curve Verification Pipeline collage

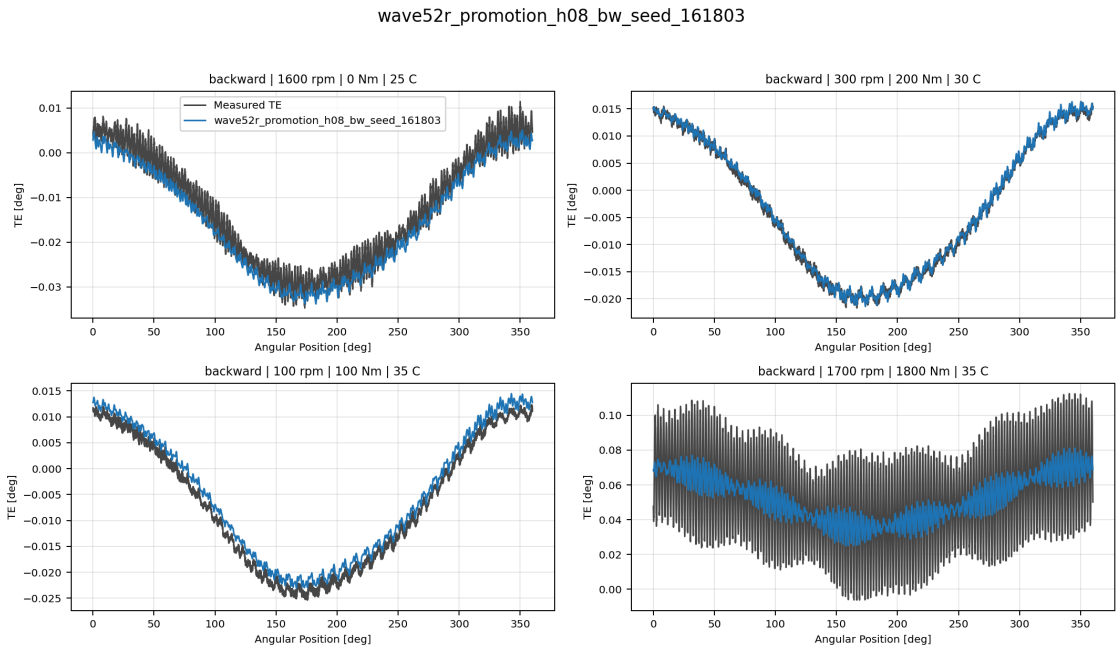
Collage Gallery - Backward Wave52r Offline Leader Cross Surface Promotion Models

wave52r_promotion_k01_bw_seed_271828



wave52r_promotion_k01_bw_seed_271828 TE Curve Verification Pipeline collage

wave52r_promotion_h08_bw_seed_161803



wave52r_promotion_h08_bw_seed_161803 TE Curve Verification Pipeline collage

Output Artifacts

- output directory:
output\analysis\wave_5_2r\offline_leader_cross_surface_track2\visual_collages\backward\2026-07-31-13-38-48__track2_best_model_collage_report ;
- summary YAML:
output\analysis\wave_5_2r\offline_leader_cross_surface_track2\visual_collages\backward\2026-07-31-13-38-48__track2_best_model_collage_report\track2_best_model_collage_summary.yaml ;
- metrics CSV:
output\analysis\wave_5_2r\offline_leader_cross_surface_track2\visual_collages\backward\2026-07-31-13-38-48__track2_best_model_collage_report\track2_best_model_collage_metrics.csv ;
- report Markdown:
doc\reports\analysis\model_development_waves\wave_5_2\offline_leader_global_promotion\visual_collages\backward\[2026-07-31]\track2_best_model_collage_report.md .